Abhishek Garia

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Education

Guru Tegh Bahadur Institute Of Technology 2021 – 2025
Bachelor of Technology *CGPA:9.189*

Ryan International School 2020 –2021
Senior Secondary(CBSE board)

Experience

Data Science and Research Methodologies July 2023 – Sep. 2023
Shaheed Rajguru College Of Applied Sciences *Delhi, India*

- Engaged with mentors, professionals, and peers, fostering a collaborative learning environment.
- Went into machine learning algorithms, statistical modeling, and predictive analytics.
- Learned the subtleties of experimental design, hypothesis testing, and the central role of Python in automation tasks and streamlining research workflows.

Projects

Sign Language Detection | *Python, Pandas, OpenCV, Mediapipe, Matplotlib, Keras, Tensorflow* Link

- Developed a sign language detection model using Python libraries, including OpenCV and TensorFlow.
- Implemented an LSTM model for the detection of Indian Sign Language gestures in real time.
- Achieved 90% accuracy in detecting Indian Sign Language gestures in real-time, benefiting users with hearing impairments.

Drowsiness Detection | *Python, Pandas, Numpy, Scikit Learn, Tensorflow, Keras, PyTorch, Open CV* Link

- Developed an object detection model to identify and determine whether a person is drowsy or not in images and video feeds using YOLOv5 and RCNN.
- Collected a diversified set of images and videos to create a custom dataset and manually labeled them using labelImg.
- The model has been evaluated with metrics like precision, recall, mean Average Precision.

Text Generation Using RNN |Python, Pandas, Numpy, Tensorflow, Keras, Scikit Learn Link

- Developed a text generation model using Recurrent Neural Networks (RNN) to generate contextually relevant text.
- Used Python libraries like TensorFlow and Keras to create and train the RNN model for text generation.
- Improved model performance by incorporating data preprocessing and vectorization techniques.

Vision Transformer | *Python, Pandas, Numpy, Scikit Learn, Keras* Link

- Applied the Vision Transformers (ViTs) to achieve better performance in classifying images on the CIFAR-10 dataset.
- Utilized self-attention mechanisms natively provided in Vision Transformers to capture long-range dependencies within image data for better classification accuracy.
- Analyzed performance differences between Vision Transformers and traditional convolutional neural networks (CNNs).
- Identifying key advantages led to a 15% overall image classification accuracy increase on the CIFAR-10 dataset.

Region Analysis and Object Replacement | *Python, SAM2, Stable Diffusion, Pytorch, EasyOCR* Link

- Worked with collaborators to implement SAM for video object segmentation with 85% recognition accuracy.
- Clearly communicated technical insights to peers, which helped integrate Stable Diffusion for seamless inpainting.
- Experimenting Latent Diffusion models, including VideoLDM, towards achieving 20% improvement in frame coherence and quality of object replacement.

Certificates

Internship in Python for Data Science and Research Methodologies 2023
Shaheed Rajguru College Of Applied Sciences For Women, Polytechnic Institute of Portalegre

Machine Learning A-Z: AI, Python & R 2024
Online course - Udemy

Technical Skills

Languages: Java, Python, C, SQL ,R

Libraries: Pandas, NumPy, Matplotlib, Scikit Learn, Natural Language Toolkit, Seaborn, Tensorflow, PyTorch, OpenCV, Mediapipe, Streamlit

Developer Tools: Git, Google Cloud Platform, VS Code, Visual Studio